

Nikhil Manikanta Vallabhaneni

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PROFESSIONAL SUMMARY

Computer Science Engineering student at Lovely Professional University with hands-on experience in machine learning, Python-based web development, and data analytics. Delivered end-to-end ML projects achieving predictive accuracy improvements using Scikit-Learn and Django. Certified in Oracle Data Platform, GenAI-powered analytics (TCS/Forge), and Deloitte Technology simulation. Strong foundation in data structures, algorithms, and AI-driven solutions.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, HTML, CSS (Bootstrap)

Frameworks & Libraries: Django, Scikit-Learn

Machine Learning: Regression, Classification, Feature Engineering

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, VMware, Streamlit

Cloud & Databases: Oracle Data Platform (OCI), SQL

Soft Skills: Problem-Solving, Project Management, Team Collaboration, Adaptability

EXPERIENCE

Machine Learning Intern

Feb 2026 – Apr 2026

1Stop.ai [View Certificate]

Remote

- Developed and evaluated machine learning models for predictive analytics across healthcare, demand forecasting, and activity recognition domains.
- Implemented algorithms including K-Nearest Neighbors (KNN), Decision Trees, Multi-Layer Perceptron (MLP), and LightGBM with a focus on data preprocessing and feature engineering.
- Performed end-to-end ML workflows including data cleaning, feature selection, model training, hyperparameter tuning, and performance evaluation.
- Gained practical experience in model development, evaluation, and documentation across multiple real-world datasets.

PROJECTS

Diabetes Prediction System | *Python, Django, Scikit-Learn, HTML/CSS/JS, Bootstrap*

Aug 2025 – Nov 2026

- Built a full-stack ML web application using Django that predicts diabetes risk in real time, processing user inputs through a trained classification model with approximately 85% accuracy.
- Integrated a Scikit-Learn ML model with a Django REST backend, enabling instant browser-based predictions and reducing manual health screening time.
- Designed a mobile-responsive frontend with Bootstrap and JavaScript, improving user accessibility across devices.

Bike Sharing Demand Prediction | *Python, Pandas, Scikit-Learn, RFE, VIF*

Jan 2025 – Apr 2025

- Built a regression pipeline to predict bike rental demand from time-series and environmental features, achieving an R^2 score of 0.82 on test data using optimized linear regression models.
- Performed data cleaning, feature engineering, and multicollinearity reduction via Recursive Feature Elimination (RFE) and Variance Inflation Factor (VIF), reducing irrelevant features by 40% and improving model generalization.
- Evaluated multiple models using R^2 , MAE, and RMSE metrics, selecting the best-fit model for production-ready deployment.

Human Activity Recognition | *Python, Scikit-Learn, Pandas, Sensor Data*

2025

- Developed a classification pipeline to recognize six human activities from time-series IMU sensor data, achieving over 90% accuracy through parameter-tuned models.
- Applied preprocessing and feature extraction techniques on raw sensor signals, reducing noise and improving pattern separability across activity classes.
- Evaluated classifier effectiveness using accuracy scores and confusion matrices, identifying and resolving class imbalance issues to ensure robust multi-class performance.

CERTIFICATIONS

Oracle Data Platform 2025 Certified Foundations Associate <i>Oracle University Valid until Apr 2028</i> Covers data management, cloud data platforms, and Oracle infrastructure fundamentals.	Apr 2026
GenAI Powered Data Analytics Job Simulation <i>TCS / Forage</i> Completed practical tasks in exploratory data analysis, delinquency prediction with AI, data storytelling, and implementing AI-driven business strategies.	Apr 2026
Technology Job Simulation <i>Deloitte / Forage</i> Completed industry-simulated tasks in coding and software development aligned with Deloitte’s technology consulting practice.	Apr 2026
ChatGPT-4 Prompt Engineering: Generative AI & LLMs <i>Udemy</i>	May 2024
Build Generative AI Apps with No-Code Tools <i>Online Platform</i>	Apr 2024

TRAINING

CodeSmart Foundation <i>C, C++ and Data Structures</i> - Completed structured training in C/C++ covering pointers, memory management, OOP principles, and core data structures including arrays, linked lists, stacks, queues, and trees. - Solved 30+ coding exercises and mini-assignments focused on algorithmic reasoning and code correctness.	Jun 2025 – Jul 2025
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ACHIEVEMENTS

- Top 100** – YoungTurks24, Naukri Campus (Sep 2024)
- Top 10** – Tech Club Competition, NewTalent (Feb 2025)

EDUCATION

Lovely Professional University <i>Bachelor of Technology in Computer Science and Engineering – CGPA: 7.51 / 10.0</i> Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Management, Operating Systems, Web Development	Aug 2023 – Present <i>Phagwara, Punjab</i>
Sasi Junior College <i>Intermediate – PCM (Physics, Chemistry, Mathematics) – Percentage: 97%</i>	Mar 2022 – May 2023 <i>Velivennu, Andhra Pradesh</i>